

Kano Kit

WHAT'S IN THE KIT

- 1 Kano computer kit with step-by-step build book



QUICK-START GUIDE

1. Follow the illustrated book to assemble the computer.
2. Boot up and complete the on-screen setup story.
3. Work through Kano's coding challenges and apps.
4. Connect to a monitor or TV via HDMI.

FULL LESSONS IN THIS GUIDE

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PRO TIP: One kit = one small group at a time; works well as a rotating 'engineer of the week' station.

Kano Kit

LESSON · GRADES 3-5

Build the Machine

LEARNING OUTCOMES

- Students will assemble a computer from its major components
- Students will name each component's job in the system
- Students will present a technical process to an audience

MATERIALS

- Kano kit with build book
- Monitor/TV with HDMI
- Component name cards for labeling

INSTRUCTIONS

1. Set the station with the build book open and parts laid out.
2. A small crew assembles step-by-step, saying each part's name and job aloud.
3. Attach component name cards as each piece goes in.
4. Boot up — the working screen is the payoff.
5. The crew demos 'their' computer to the class, explaining what's inside.
6. Rotate a new crew next session for teardown and rebuild.

ACTIVITY

Build the Machine — assemble a real computer, then present it. Naming each part's job while placing it builds a mental model of every device students touch.

Kano Kit

LESSON · GRADES 6-8

Hardware Anatomy Lab

LEARNING OUTCOMES

- Students will identify computer hardware components and their functions
- Students will compare architecture across different devices
- Students will diagram a computer system accurately

MATERIALS

- Kano kit with build book
- A school Chromebook (for comparison, not disassembly)
- Diagram template: CPU, memory, storage, I/O, power

INSTRUCTIONS

1. Station crew assembles the Kano while completing the diagram template.
2. For each Kano part, find its equivalent in the Chromebook spec sheet.
3. Build a comparison table: Kano vs. Chromebook vs. their phone.
4. Discuss trade-offs: why is one upgradeable and another sealed?
5. Boot and verify, then connect diagrams to the running system.
6. Exit ticket: trace what happens between pressing power and seeing the desktop.

ACTIVITY

Hardware Anatomy Lab — build the Kano, diagram its organs, and map them onto the sealed devices students use daily. Computer architecture becomes something they've held.